Evaluating Birth Time’s Effect on Education and Earning

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Research Motivation

- Workers' earnings increase as educational attainment rises. (BLS)
- Education years vary by QUARTER of BIRTH within one year due to some state laws' minimum drop out age
- Indirectly affect return on education.

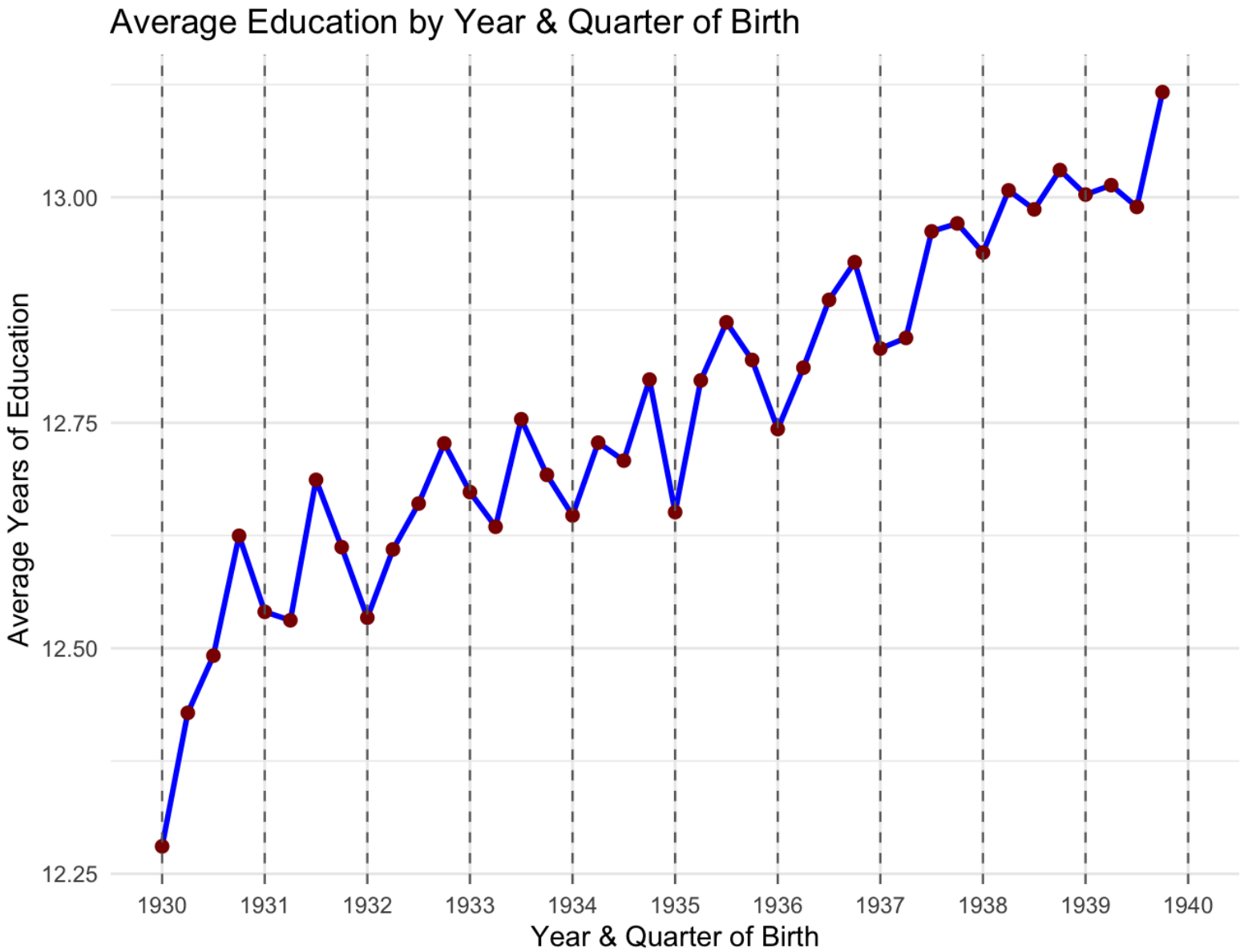
Data Description

The database consists of information of 329,509 individuals born between 1930 and 1939

10 Variables:

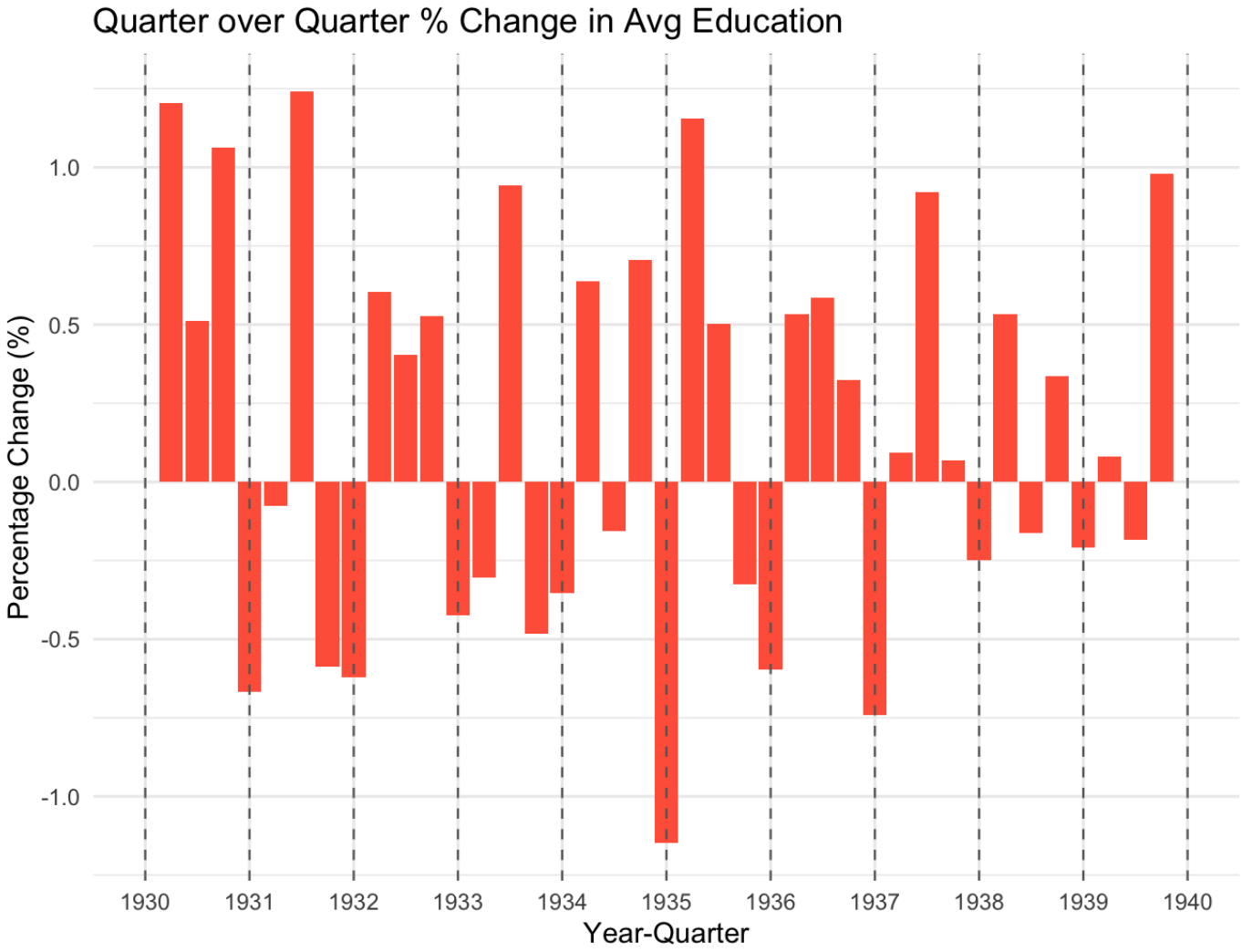
Age, **Completed Years of Education**, $\mathbf{log}(\boldsymbol{Wage})$, Marital Status, State of Residence, Race (Black or not), Metro Area Resident Status (Living in a metro area or not), **Quarter of Birth**, **Year of Birth**, and Region of Residence

Avg. Years of Completed Education of People Born in Every Year-Quarter



Seasonal Pattern:

People born in Q1 of a year is always less educated (on average) compared with people born in Q4 of the previous year



A summary of the Previous Graph

Cohort (by Birth Quarter)	Occurrence of Negative Percentage Change	Total Occurrence	Pr [Cohort less educated than previous-quarter-born cohort]
1	9	9	100%
2	2	10	20%
3	3	10	30%
4	3	10	30%

Estimating Year & Quarter of Birth's Effect on Completed Education

Transform year of birth:

$$\boldsymbol{Centered\ YoB = YoB\ - Mean(YoB)}$$

Regression:

$$\boldsymbol{Edu_i = \beta_0 + \beta_1 * (Centered\ YoB)_i + \beta_2 * qob_{1_i} + \beta_3 * qob_{2_i} + \beta_4 * qob_{3_i} + \varepsilon_i}$$

People born in Q4 is the reference group

Intercept(β_0): estimated education level for people born in Q4 in the middle-year

β_1:estimated change in education level for one later year of birth

$\beta_2, \beta_3, \beta_4$: Difference in education level of people born in Q1, Q2, Q3, compared with Q4

Estimated Coefficients:

	Dependent variable:
	edu
yob_centered	0.060***
	(0.002)
qob_1	-0.148***
	(0.016)
qob_2	-0.092***
	(0.016)
qob_3	-0.036**
	(0.016)
Constant	12.838***
	(0.012)
Observations	329,509
R2	0.003
Adjusted R2	0.003
Residual Std. Error	3.276 (df = 329504)
F Statistic	256.056*** (df = 4; 329504)
Note:	*p<0.1; **p<0.05; ***p<0.01

Two-Stage Least Squares (stage 1)

$$E_i = \beta_1 Q_{ij} + \Sigma\beta_k Y_{ik} + \Sigma\beta_z S_{iz} + \Sigma\beta_i V_i + \epsilon_i$$

$$lnW_i = \beta_2 E_i + \beta_3 A_i + \beta_4 M_i + \beta_5 B_i + \Sigma\omega_k Y_{ik} + \Sigma\gamma_z S_{iz} + \Sigma\beta_n R_{in} + \xi_i$$

- E_i *indicates years of education for person* i
- Q_{ij} *is a dummy variable indicating the quarter of birth*$(= 1$ *if quarter* $1, = 0$ *otherwise*$)$
- Y_{ik} *is the year of birth* k
- S_{iz} *is the state* z
- V_i *represents other independent variables that will affect years of education*

Two-Stage Least Squares (stage 2)

$$E_i = \beta_1 Q_{ij} + \Sigma\beta_k Y_{ik} + \Sigma\beta_z S_{iz} + \Sigma\beta_i V_i + \epsilon_i$$

$$lnW_i = \beta_2 E_i + \beta_3 A_i + \beta_4 M_i + \beta_5 B_i + \Sigma\omega_k Y_{ik} + \Sigma\gamma_z S_{iz} + \Sigma\beta_n R_{in} + \xi_i$$

- W_i *is the wage indicator*
- A_i *is a dummy variable indicating whether person i lives in metropolitan area* $(= 1\ if\ yes)$
- M_i *is a dummy variable indicating whether person i is married* $(= 1\ if\ yes)$
- B_i *is a dummy variable indicating the race of person i* $(= 1\ if\ black, = 0\ otherwise)$
- R_{in} *is the region*

OLS VS. TSLS

OLS and TSLS Estimates of the Return to Education

	(1)	(2)	(3)	(4)
INDEPENDENT VARIABLE	**OLS**	**TSLS**	**OLS**	**TSLS**
Years of Education	**0.067 (***)**	**0.104 (***)**	**0.063 (***)**	**0.096 (***)**
Metropolitan Area (1 = yes)	**---**	**---**	**0.171 (***)**	**0.139 (***)**
Married (1 = yes)	**---**	**---**	**0.249 (***)**	**0.240 (***)**
Race (1 = black, 0 = non-black)	**---**	**---**	**-0.255 (***)**	**-0.220 (***)**
Years of Birth	**YES**	**YES**	**YES**	**YES**
State	**YES**	**YES**	**YES**	**YES**
Region	**---**	**---**	**YES**	**YES**

significance level in parentheses

Conclusion

- People born in Q1 receive less education on average, because they reach legal dropout age earlier than the others.
- Years of education increases personal earnings in the future.
- Metropolitan residence, marriage and race play significant roles too.

- All in all, EDUCATION MATTERS!